Nhill 2.1 — Installation guide

Français : ../fr/guide-installation.md

This guide covers installing the **applications** (Linux, Windows, Android), the **server** (Linux) and upgrading from 2.0 or 0.1.0. Building the packages: build-guide.md. Settings: configuration-guide.md. Marketing website: `website/README.en.md`.

> **State of the deliverables.** Every file shipped in `release/V<version>/` was actually built; a missing platform is listed in `release/V<version>/checksums/MISSING.txt`. The validation report (validation-report.md) says what was tested and on which machine. The Windows installers and the APK are **not signed** with a publisher certificate until the creators acquire one.

1. Verify a downloaded file

Before installing, compare the file's SHA-256 digest with the one in `checksums/SHA256SUMS` (published separately from the file, for example on the website):

```
sha256sum -c SHA256SUMS --ignore-missing          # Linux
```

```
Get-FileHash .\nhill-business-2.1.0-setup.exe -Algorithm SHA256   # Windows: compare by eye
```

If the digest differs, do not install the file.

2. Minimum configuration

Component	System	Memory	Notes
Nhill Business, POS, Console (desktop)	Windows 10/11 64-bit; Debian 13 or Ubuntu 24.04 (amd64)	to be measured on the target computer	1280×720 screen or larger recommended; POS works from 860×560
Nhill Business (Android)	Android 7.0 (API 24) or later, arm64 processor	—	Internet connection required
Server	Debian 13 or Ubuntu 24.04, amd64	starting point: 2 vCPU, 4 GB, 40 GB SSD	PostgreSQL 15+ (Supabase or self-hosted), synchronised clock (NTP)

The client computers' memory needs have not been measured: test on the hardware actually used (see the validation report).

3. Applications

3.1 Linux (Debian 13, Ubuntu 24.04)

Packages are built **for one specific distribution**: take those from your system's folder (`linux/debian-13/` or `linux/ubuntu-24.04/`).

```
sudo apt install ./nhill-business_2.1.0-1_amd64.deb ./nhill-pos_2.1.0-1_amd64.deb
sudo apt install ./nhill-console_2.1.0-1_amd64.deb     # for the creators of Nhill only
```

`apt` installs the required libraries from the official repositories. The applications appear in the menu (Office). Remembered sessions need a Secret Service keychain (GNOME Keyring or KWallet), present on usual desktops.

Removal: `sudo apt remove nhill-business nhill-pos nhill-console`.

3.2 Windows 10/11

1. Check the digest (§ 1), then run `nhill-business-2.1.0-setup.exe` (or `nhill-pos-…`, `nhill-console-…`). Installation asks for administrator rights (it installs into `Program Files`).
2. As long as the installer is unsigned, Windows SmartScreen may show “Windows protected your PC”. Bypass this warning **for this file only** and **only if the digest matches**: “More info” → “Run anyway”. Never turn off SmartScreen or the antivirus.
3. Start-menu shortcuts are created; remove through Settings → Apps.

Silent installation (fleet of computers): `nhill-pos-2.1.0-setup.exe /S`.

3.3 Android

1. Check the digest of `nhill-business-2.1.0-arm64-v8a.apk`.
2. Open the file on the phone; Android asks you to allow **this app** (file manager or browser) to install apps. Allow it for this installation, then withdraw the permission (Settings → Apps → Special access → Install unknown apps).
3. An APK signed with the **debug** key (built without a release key) can be installed for trials but must not be distributed to customers; it cannot be upgraded by an APK signed with the release key (uninstall first).

Nhill POS and Nhill Console are not offered on Android (desktop applications).

3.4 First start

The server address is compiled into the application (`NHILL_SERVER_URL`). To try another server, set the `NHILL_SERVER` environment variable (desktop only). Then: user guide, Part A § 1.

4. Server (production)

Target: a Linux server for the API, a PostgreSQL 15+ database (**Supabase** or self-hosted), a domain name and a TLS proxy (Caddy). The full procedure has not yet been run on a public production server with TLS: run it on a staging server first.

4.1 Prerequisites

- Synchronised clock (NTP) — essential for TOTP and webhooks.
- Firewall: only 22 (administration, restricted) and 80/443 (Caddy) open.
- DNS: `api.<your-domain>` → server address.

4.2 Package

```
sudo apt install ./nhill-server_2.1.0-1_amd64.deb
curl -s http://127.0.0.1:8080/health     # {"database":false,"status":"setup"}
```

The package creates the system users `nhill` and `nhill-backup`, the file `/etc/nhill/server.env` (mode 600) with an `NH_SECRET_KEY` **generated on the server** and no database, then starts the service in **setup mode**. Copy `NH_SECRET_KEY` to a vault immediately (without it, establishment codes and the administrators’ TOTP secrets become unreadable).

4.3 Database

- **Supabase**: Connect → **Session pooler** (port 5432, user `postgres.<reference>`); not the *transaction pooler* (6543). No API key (`anon`, `service_role`) is needed or accepted; Nhill creates a `nhill` schema closed to Supabase's API roles.
- **Self-hosted PostgreSQL**: empty `nhill` database, an account able to create roles, `scram-sha-256` in `pg_hba.conf`, TLS if the database is remote.

4.4 Installation from Nhill Console

1. Setup token (60 minutes, single use): `journalctl -u nhill-server | grep -i -A2 "setup mode"`.
2. Nhill Console → **Install a new server** → address and token → database type, host, port, user, password → **Test the connection** → first administrator → **Initialise the database**.
3. The server creates `nhill_owner`, `nhill_app`, `nhill_backup` (random passwords), applies the migrations, checks isolation, writes `/etc/nhill/server.env` and restarts. The database administration password is not kept; the backup account's password is shown **only once**.

Check: `/health` → `{"database":true,"status":"ok"}`. Command line: `NH_SETUP_DB_PASSWORD=… sudo -u nhill nhill-server --config /etc/nhill/server.env setup --host … --user … --admin-email … --admin-name "…"`.

4.5 TLS and settings

```
sudo apt install caddy
sudo cp /usr/share/nhill/examples/Caddyfile.example /etc/caddy/Caddyfile   # replace
api.example.com
sudo systemctl reload caddy
curl -sI https://api.<domain>/health       # 200 + Strict-Transport-Security
```

In `/etc/nhill/server.env`: `NH_PUBLIC_BASE_URL=https://api.<domain>`, `NH_TRUST_PROXY=1`; then `sudo systemctl restart nhill-server` and `sudo -u nhill nhill-server --config /etc/nhill/server.env check`.

4.6 Backups, first access, plans

- Encrypted backups (age): `docs/sauvegarde-restauration.md` (French); `sudo systemctl enable --now nhill-backup.timer`, then **a restore test**.
- Nhill Console → **First access** (30-minute token, password of at least 8 characters, mandatory TOTP).
- **Plans and prices**: no price is preconfigured.
- **Mail** (2.1): SMTP relay with TLS (`NH_EMAIL_TRANSPORT=smtp`, `NH_SMTP_*`); without it, owners and administrators cannot recover a forgotten password.
- **Payment providers**: stay in test mode until a successful test payment with the provider's account.

5. Upgrading to 2.1.0

From 2.0.x:

1. **Back up** the database: `sudo systemctl start nhill-backup.service`.
2. Server: `sudo apt install ./nhill-server_2.1.0-1_amd64.deb`; at start-up the service applies migration **0011** (forgotten password: two columns in `users`, the `password_resets` table, three platform settings). Schema: 0001–0011.

3. Mail: set `NH_EMAIL_TRANSPORT=smtp` and the `NH_SMTP_*` variables (configuration-guide.md § 1 and § 5), then `sudo -u nhill nhill-server --config /etc/nhill/server.env check` and try “Forgot your password?” with a test account.
4. Applications: install the new versions **before** using the reset by an administrator. A 2.0 client is still accepted by the 2.1 server (API `v1`), but it cannot show the required change of a temporary password: the member could not sign in from that device.
5. Remembered session: a user whose session was already kept by 2.0 (Linux) keeps it; elsewhere, “Remember me” takes effect at the next sign-in.

From 0.1.0: same steps; 2.0 added no migration (the 8-character rule applies to new passwords, with no forced change for existing ones). Upgrade server and clients together (0.1.x clients reject passwords shorter than 10 characters locally).

Rollback to 2.0: restoring the backup from step 1 is the safe method. Reinstalling only the 2.0 package leaves migration 0011 in place (the 2.0 server ignores the added columns and table); this case has not been tested, and a temporary password already given would work there as an ordinary password.

6. Troubleshooting

Symptom	Action
“Server problem” in the application	`curl https://api.<domain>/health` ; `journalctl -u nhill-server`
Session not remembered on Linux	install/unlock GNOME Keyring or KWallet
Session not remembered on Windows or Android	check that “Remember me” is ticked (Settings › My account › This device); Android: set a screen lock; a “secure storage unavailable” message appears if the system refuses to save
“Forgot your password?”: no code received	server: `NH_EMAIL_TRANSPORT=smtp` ? relay error in `journalctl -u nhill-server` ; user: spam folder; an employee of a single establishment receives the instruction to contact their administrator, not a code
“Update required”	install the version requested by the server
Android: “App not installed”	an APK with another signature is present: uninstall it
Windows: application does not start	reinstall; check that the antivirus has not quarantined a file
Server refuses to start, `server.env` readable	`chmod 600 /etc/nhill/server.env`
Server: modified migration refused	never edit a published migration; restore the original